

NASRALLAH RAFI

Electrical Engineer

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Profile

Electrical Engineer specializing in Embedded Systems, Battery Management Systems (BMS), and Intelligent Energy Storage, with a research focus on LiFePO battery modeling and AI-assisted state estimation. Experienced in industrial BMS hardware design, physical battery modeling and Equivalent Circuit Models (ECM), Battery Digital Twins, and advanced SOC estimation using Unscented Kalman Filtering (UKF) and LSTM-based residual correction for electric mobility applications

Education

ENSET, Mohammedia – State Engineer's Degree in Electrical Engineering and Industrial Systems Control 2023 – 2026

Faculty of Sciences Ben M'sik, Casablanca – Bachelor's Degree in Physical Sciences (Electronics) 2022 – 2023

Faculty of Sciences Ben M'sik, Casablanca – DEUG in Physics 2020 – 2022

Technical High School, Mohammedia – Baccalaureate in Electrical Sciences and Technologies 2019 – 2020

Professional Experience

End-of-Studies Engineering Internship RD – Emove Véhicules Feb 2026 – May 2026

Project: Systemic resolution of a critical premature shutdown defect and energy gauge instability during abrupt accelerations on electric scooters through a coupled Hardware / Software / Process approach.

- **Digital Twin:** Developed a high-fidelity virtual battery LFP chemistry integrating dynamic hysteresis modeling and a digital safety architecture aligned with IEC 62133 / ISO 26262 principles.
- **Estimation Algorithms & AI:** Designed and implemented a hybrid SoC estimation architecture combining a 3RC Equivalent Circuit Model (ECM), Unscented Kalman Filter (UKF), and LSTM neural network, with an AI-based residual observer.
- **Advanced Energy Management:** Integrated a bidirectional active balancing system and developed a Finite State Machine (FSM) safety supervisor for degraded-mode management and system protection.
- **Hardware & Embedded Design:** Designed and routed a 16S2P BMS board in Altium Designer based on an STM32H7 microcontroller, integrating EMC considerations, power protections, and I2C, SPI, and CAN interfaces.
- **Validation & Robustness:** Validated SoC estimation algorithms through MIL simulation on the digital twin (RMSE < 2%) and experimental HiL validation using an instrumented prototype with Arduino and high-side MOSFET switching.
- **Quality Management (QMS):** Developed and implemented an industrial cell qualification procedure aligned with ISO 9001 quality-management principles.
- **Supervision & Digitalization:** Developed a real-time industrial dashboard using MQTT and REST API for battery-pack traceability, production monitoring, and KPI analysis.

Engineering Internship – SMCV Jul 2025 – Sep 2025

- Requirement analysis, programming, and supervision of a plastic injection machine (TIA Portal).
- Comprehensive design, study, and automation of a hot nut insertion press, optimizing productivity and process repeatability.

Technical Internship – ONEE Jul 2024 – Aug 2024

- Analysis of Siemens digital relays and development of testing procedures.

Patents

Automated System and Process for Welding and Assembling Corrugated Web Beam Profiles

Industrial Patent – Lead Inventor – Issued Nov – 2025 OMPIC No. 65133

Technical Skills

Expertise: Embedded Systems & IoT, Battery Management Systems (BMS), Power Electronics, Industrial Automation, Model-Based Design (MBD), Functional Safety.

Software: Altium Designer, MATLAB/Simulink, Siemens TIA Portal, PLCnext Engineer, Caneco BT, ETAP, Proteus, Ansys (HFSS/Mechanical), Autodesk Inventor, Fusion 360.

Programming: C/C++, Python, Ladder/SCL | RTOS/FreeRTOS, CAN Bus, I2C, SPI, UART, MQTT.

Languages: Arabic (Native), English (C2 – Advanced), French (B2 – Intermediate).